

Independent Cost Function Model Predictive Current Control for SMPMSM Drive

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Abstract—To achieve controllable d -axis and q -axis current performance with inverter switching frequency reduction, an independent cost function model predictive current control (ICF-MPC) is proposed for surface mounted permanent magnet synchronous motor (SMPMSM) drives. First, two independent cost functions are designed for d -axis and q -axis individual current error. Then, a rectangular constraint that satisfies dq -axes current performance is presented to obtain the voltage vector space. Subsequently, a direct selection and calculation method is proposed to obtain the reference voltage vector, based on the principle of balancing current ripple and switching frequency. The proposed ICF-MPC achieves simultaneous optimization of d -axis current performance, q -axis current performance, and switching frequency without weighting factors. The controllable current performance and switching frequency reduction are demonstrated by the experimental results.

Index Terms—Independent cost function (ICF), permanent magnet synchronous motor (PMSM), predictive control, switching frequency.

I. INTRODUCTION

MODEL predictive control (MPC) is widely used in permanent magnet synchronous motor (PMSM) drives due to its simple structure, high dynamic response, and multiobjective optimization. To improve the efficiency of PMSM drives, it is crucial to minimize the power losses of the inverter. Additionally, reducing power losses not only eases the burden on the cooling system but also helps to prevent overheating of the inverter and associated electronic components, thereby enhancing the reliability of PMSM drives. The power losses of power semiconductor devices mainly include conduction losses and switching losses. Among them, switching losses are directly related to the inverter switching frequency. Therefore, reducing

the switching frequency is one of the important means to optimize inverter power losses.

Generally, the current ripple and the inverter switching frequency are two conflicting and different-scale optimization objectives for MPC. Researchers have shown interest in achieving optimal control for multiobjectives. To achieve multiobjective optimization in MPC, a most common method is to introduce weighting factors into the cost function. However, tuning the weighting factors is time-consuming. Optimizing these factors for ideal control performance under various working conditions can be challenging. Furthermore, an improper weighting factor could result in undesirable system behavior [7]. To overcome this drawback, various methods have been proposed to optimize the weighting factors, both online and offline, such as empirical method [1], genetic algorithm [2], artificial neural network [3], [4], particle swarm optimization [5], and fuzzy control [6].

Additionally, certain methods have been proposed to reduce the number of weighting factors. In [8], torque and stator flux are unified into voltage vector control, and the weighting factors are only introduced into the neutral point voltage and the switching frequency terms. In [9], the influence of the weighting factors of various terms in the cost function on different control indicators is analyzed. This facilitates the tuning of the weighting factors. In [10], the weighting factor is only associated with the neutral point potential term in the cost function, and the switching frequency term is regarded as a hard switch constraint. Moreover, the voltage vector that meets the number of switching transitions constraint will continue to be used in the next control period. This extends the application time of the same voltage vector and reduces the number of switching transitions in power semiconductor devices. A similar method is also employed in [11]. In [12], the active and reactive power of the grid-connected inverter is predicted using a finite control set model predictive control (FCS-MPC) method. The basic voltage vector that minimizes the cost function for the next two steps is selected, and applied in the next two steps to improve the current control accuracy and reduce the switching frequency. In [13], a hybrid modulation method based on FCS-MPC is proposed to reduce the inverter switching loss by selecting the applied sequence that minimizes the cost function relation to inverter switching loss. In addition, similar ideas are adopted in [14] and [15]. Recently, researchers have used selective harmonic elimination pulse width modulation (SHE-PWM) methods to enhance the current quality of PMSM drives [16] and inverters [17] and reduce the switching frequency of inverters. However, compared to traditional methods, their computational burdens are significantly increased.

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Some researchers have proposed MPC methods that focus on multiobjective optimization without weighting factors. In [18], a predictive torque control method is proposed, which transforms torque and stator flux into stator flux control to eliminate the weighting factors. In [19], the second voltage vector at the current step is used as the first voltage vector at the next step to reduce the switching frequency. However, this approach limits the selection of the voltage vector, which affects the steady-state control performance of stator flux. To improve the steady-state torque and stator flux control performance, two cost functions are designed for model predictive torque control (MPTC) in [20]: one for torque and the other for stator flux. However, the sequence evaluation scheme in the method focuses only on either the control performance of torque or stator flux linkage, which can reduce the overall system freedom. Consequently, this can lead to higher steady-state current ripple when minimizing the torque and stator flux sequence. Additionally, improving the current control performance using the method in [20] requires an increase in the switching frequency as a trade-OFF [21]. In [22], the top-three voltage vectors that correspond to the minimized torque and flux objective cost functions are considered individually, and the optimal one is selected to minimize both torque and flux ripples. To more directly optimize the switching frequency, two cost functions related to stator flux and switching frequency are established in [23]. Then, each available voltage vector is evaluated and ranked. The voltage vector with the lowest average rank is selected as the reference voltage vector.

Some of the methods previously mentioned rely on weighting factors, while others avoid the use of the weighting factors but are still single-vector based FCS-MPC methods. Thus, the steady-state performance is inevitably subject to degradation. Moreover, traditional MPC methods use the sum of the dq -axes current errors as the cost function, resulting in optimization of only the cumulative dq -axes current ripples. Therefore, achieving the multiobjective optimization of independent dq -axes current performance, and inverter switching frequency reduction of PMSM drive without weighting factors has become a challenge. In the proposed ICF-MPC, two independent cost functions of d -axis and q -axis current error are initially presented. By setting the dq -axes current constraint, the proposed ICF-MPC can extend the voltage vector space, represented as a rectangular constraint within the voltage hexagon. Then, the reference voltage vector, which can simultaneously optimize both the switching frequency and the current ripple, is analytically obtained. This method proposes the concept of dq -axes current constraint to achieve the independent and controllable dq -axes current performance according to the specification of requirements. Moreover, the switching frequency can be effectively reduced. Finally, the effectiveness of the proposed method has been verified through experiments.

II. MATHEMATICAL MODEL OF SMPMSM

Considering the parametric uncertainties and the inverter nonlinearity, the mathematical model of the surface mounted

permanent magnet synchronous motor in the synchronous rotating coordinate system can be defined as follows:

$$\begin{cases} \frac{di_d}{dt} = \frac{1}{L_s} u_d^* - \frac{R_s}{L_s} i_d + \omega_e i_q - \frac{1}{L_s} \Delta u_d \\ \frac{di_q}{dt} = \frac{1}{L_s} u_q^* - \frac{R_s}{L_s} i_q - \frac{1}{L_s} \omega_e (L_s i_d + \varphi_f) - \frac{1}{L_s} \Delta u_q \end{cases} \quad (1)$$

where u_d^* and u_q^* are the dq -axes reference voltages. i_d and i_q are the dq -axes currents. R_s , L_s , and φ_f denote the nominal parameters of stator resistance, stator inductance, and permanent magnet flux linkage, respectively. ω_e is the electric angular speed of the motor rotor. Δu_d and Δu_q are the dq -axes disturbances caused by parametric uncertainties and inverter nonlinearity.

To avoid the negative effects of the parametric uncertainties and the inverter nonlinearity, the ultra-local model [24] of SMPMSM is used in this study as follows:

$$\begin{cases} \frac{di_d}{dt} = F_d + \alpha_d u_d^* \\ \frac{di_q}{dt} = F_q + \alpha_q u_q^* \end{cases} \quad (2)$$

where α_d and α_q are the scaling factors of the dq -axes reference voltages. F_d and F_q are the dq -axes known and unknown parts, including the back-electromotive force, stator resistance voltage drop and disturbances caused by the parametric uncertainties and the inverter nonlinearity. Moreover, F_d and F_q can be estimated online by the algebraic parameter identification method [24].

By using the forward Euler discretization method, the predictive model for the current of SMPMSM, based on the ultra-local model, is obtained as follows:

$$\begin{cases} i_d(k+1) = T(\hat{F}_d(k) + \alpha_d u_d^*(k-1)) + i_d(k) \\ i_q(k+1) = T(\hat{F}_q(k) + \alpha_q u_q^*(k-1)) + i_q(k) \end{cases} \quad (3)$$

where $i_d(k+1)$ and $i_q(k+1)$ represent the predicted dq -axes currents at the $(k+1)$ th control period. $i_d(k)$ and $i_q(k)$ are the measured dq -axes currents at the (k) th control period. $u_d^*(k-1)$ and $u_q^*(k-1)$ are the dq -axes reference voltages at the $(k-1)$ th control period, and are applied during the (k) th control period. $\hat{F}_d(k)$ and $\hat{F}_q(k)$ are the estimated values of F_d and F_q at the (k) th control period. T is the sampling time.

III. PRINCIPLE OF THE PROPOSED METHOD

In the section, Part A outlines the independent dq -axes current constraints. Part B illustrates the influence of the independent current constraints on the range of the rectangular constraint, and reveals the relationship between the independent current constraints and the reference voltage vector. Part C elaborates on the optimization process of the reference voltage vector.

A. Independent Current Constraints

Considering the delay in digital control systems, (3) is used to predict the dq -axes currents for the next control period, thereby compensating for the one-step delay in the control system. The

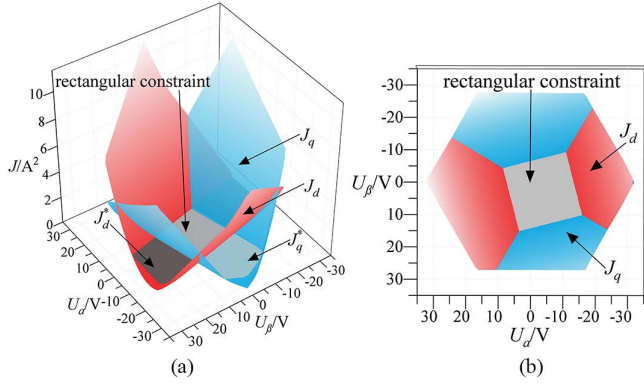


Fig. 1. Three-dimensional surface of independent cost functions. (a) Three-dimensional schematic diagram. (b) Diagram of rectangular constraint.

predicted dq -axes currents at the $(k + 2)$ th control period are shown as follows:

$$\begin{cases} i_d(k+2) = T(\hat{F}_d(k) + \alpha_d u_d^*(k)) + i_d(k+1) \\ i_q(k+2) = T(\hat{F}_q(k) + \alpha_q u_q^*(k)) + i_q(k+1) \end{cases} \quad (4)$$

where $i_d(k+2)$ and $i_q(k+2)$ are the predicted dq -axes currents at the $(k+2)$ th control period. $u_d^*(k)$ and $u_q^*(k)$ are the dq -axes reference voltages at the (k) th control period.

To achieve the individual control of the dq -axes currents, the independent cost functions are designed as follows:

$$\begin{cases} J_d = (i_d^* - i_d(k+2))^2 \\ J_q = (i_q^* - i_q(k+2))^2 \end{cases} \quad (5)$$

where J_d and J_q represent the dq -axes current cost functions; i_d^* and i_q^* denote the dq -axes reference currents.

Using the inverse Park transformation, $u_d^*(k)$ and $u_q^*(k)$ are converted into $u_\alpha^*(k)$ and $u_\beta^*(k)$

$$\begin{pmatrix} u_\alpha^*(k) \\ u_\beta^*(k) \end{pmatrix} = \begin{pmatrix} \cos \theta_r & -\sin \theta_r \\ \sin \theta_r & \cos \theta_r \end{pmatrix} \begin{pmatrix} u_d^*(k) \\ u_q^*(k) \end{pmatrix} \quad (6)$$

where θ_r denotes the rotor position of SMPMSM.

Then, by substituting (3) and (6) into (4), the predicted dq -axes currents at the $(k+2)$ th control period can be obtained and then substituted into (5). Therefore, the independent cost functions can be obtained

$$\begin{cases} J_d = (V_d - \alpha_d T(\cos \theta_r u_\alpha^*(k) + \sin \theta_r u_\beta^*(k)))^2 \\ J_q = (V_q - \alpha_q T(-\sin \theta_r u_\alpha^*(k) + \cos \theta_r u_\beta^*(k)))^2 \end{cases} \quad (7)$$

$$\text{where } \begin{cases} V_d = i_d^* - i_d(k+1) - T\hat{F}_d(k) \\ V_q = i_q^* - i_q(k+1) - T\hat{F}_q(k) \end{cases}$$

In a specific control period, substituting all voltage vectors from the voltage hexagon into (7) yields J_d and J_q . J_d for all voltage vectors in the voltage hexagon constitute the J_d surface, while J_q for all voltage vectors form the J_q surface. These surfaces are represented by the red and blue colors, respectively, in Fig. 1. In Fig. 1(a), the horizontal axes represent the α -axis voltage u_α and the β -axis voltage u_β , respectively. Furthermore, the

vertical axis indicates the results of the independent cost functions for each voltage vector in the stationary coordinate frame.

Furthermore, define the constraints of the d -axis cost function, denoted as J_d^* , which is also the d -axis current constraint and is represented by the dark grey plane in Fig. 1(a). The red surface that lies under J_d^* plane complies with the d -axis current constraint. Therefore, the projection of the dark grey plane in the voltage hexagon is the set of all voltage vectors in the voltage hexagon that satisfy the d -axis current constraint J_d^* . J_q^* has taken on a similar meaning to J_d^* , and is denoted in light grey. The blue surface below J_q^* plane satisfies the q -axis current constraint. Therefore, the projection of the light grey plane in the voltage hexagon is the set of all voltage vectors that satisfy the q -axis current constraint J_q^* . Fig. 1(b) is the projection of Fig. 1(a) onto the $u_\alpha u_\beta$ -plane. And in Fig. 1(b), the grey area is the intersection of the projections of J_d^* plane and J_q^* plane on the $u_\alpha u_\beta$ -plane, which is defined as the rectangular constraint. Therefore, any voltage vector in the rectangular constraint can satisfy both d -axis current constraint J_d^* and q -axis current constraint J_q^* . The dq -axes current constraints are defined as the independent current constraints, which reflect the magnitude of dq -axes current errors.

The q -axis current error is directly related to the magnitude of J_q^* , which in turn influences the torque ripple in SMPMSM. In practical applications, J_q^* can be set according to allowable torque ripple in the specification of requirements of SMPMSM drive. Then, J_d^* can be initially set equivalent to J_q^* , and adjusted based on the actual performance of SMPMSM drive.

In different control periods, J_d and J_q surfaces are different, which implies that Fig. 1 changes dynamically. The data used in Fig. 1 are simulated using an SMPMSM. In the simulation, the current of SMPMSM is controlled by the proposed ICF-MPC and the speed is controlled by a dynamometer at 100 rpm. The q -axis reference current is 10 A, and the d -axis reference current is 0 A. The parameter settings for SMPMSM in the simulation are the same as those in the experimental setup below.

B. The Relationship Between the Rectangular Constraint and Independent Current Constraints

Under the independent current constraints, the derivation process of the mathematical formula of the four edges of the rectangular constraint at the (k) th control period is as follows:

The independent current constraints J_d^* and J_q^* are used to replace J_d and J_q in (7). Therefore, (8) is obtained

$$\begin{cases} J_d^* = (V_d - \alpha_d T(\cos \theta_r u_\alpha^*(k) + \sin \theta_r u_\beta^*(k)))^2 \\ J_q^* = (V_q - \alpha_q T(-\sin \theta_r u_\alpha^*(k) + \cos \theta_r u_\beta^*(k)))^2 \end{cases} \quad (8)$$

Let $x = u_\alpha^*(k)$ and $y = u_\beta^*(k)$. After expanding the above equation, it can be rewritten as follows:

$$\begin{cases} E_{d1}: y = mx + p_1 \\ E_{d2}: y = mx + p_2 \\ E_{q1}: y = nx + q_1 \\ E_{q2}: y = nx + q_2 \end{cases} \quad (9)$$

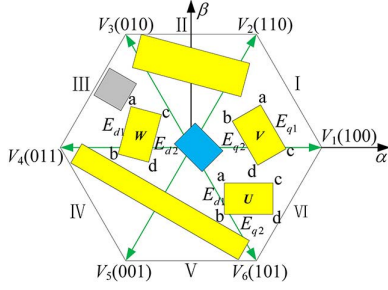


Fig. 2. Position of rectangular constraint in voltage hexagon.

where $x = u_\alpha^*(k)$, $y = u_\beta^*(k)$, $m = -(1/\tan \theta_r)$, $n = \tan \theta_r$, $p_1 = (V_d - \sqrt{J_d^*/T\alpha_d} \sin \theta_r)$, $p_2 = (V_d + \sqrt{J_d^*/T\alpha_d} \sin \theta_r)$, $q_1 = (V_q - \sqrt{J_q^*/T\alpha_q} \cos \theta_r)$, and $q_2 = (V_q + \sqrt{J_q^*/T\alpha_q} \cos \theta_r)$.

During the (k) th control period, all variables in (9) are known except $x(u_\alpha^*(k))$ and $y(u_\beta^*(k))$. Therefore, $x(u_\alpha^*(k))$ and $y(u_\beta^*(k))$ satisfy the binary linear equations in each control period. These equations are the mathematical expressions of the four edges of the rectangular constraint.

The range of the rectangular constraint is determined by the mutually independent J_d^* and J_q^* , which facilitate the individual and flexible control of dq -axes current errors. Increasing J_d^* and J_q^* correspondingly expands the range of the rectangular constraint, and provides greater freedom in selecting the reference voltage vector. Compared to traditional cost functions, the independent cost functions used in this paper can increase the degrees of freedom in constraining the dq -axes currents, leading to an expanded space for selecting the reference voltage vector and facilitating a reduction in the switching frequency.

C. Optimization of Reference Voltage Vector

To select the reference voltage vector, two criteria must be satisfied. First, the voltage vector must comply with the independent current constraints. Second, the voltage vector with fewer number of switching transitions should be chosen to reduce the switching losses. Therefore, the proposed ICF-MPC adopts a hybrid voltage vector method, which combines different types of voltage vectors, such as single-vector, double-vector, and three-vector. Moreover, the reference voltage vector is synthesized using the symmetrical discontinuous pulse width modulation (DPWM) mode, and only 000 is utilized as the zero-vector. According to the number of switching transitions of the voltage vector, the selection priority is sorted from high to low as follows: zero-vector (000), double-vector (100/010/001 + 000), double-vector (110/011/101 + 000) and three-vector. The number of switching transitions of the corresponding voltage vectors increases sequentially, being 0, 2, 4, and 4 times, respectively.

According to the position of the rectangular constraint in voltage hexagons, it can be divided into three cases: Case 1, Case 2, and Case 3. Case 1 means that the origin is inside the rectangular constraint, as shown in the blue region of Fig. 2. The zero-vector should be chosen as the reference voltage vector to minimize the number of switching transitions. Case 2 implies that the active voltage vectors (AVVs) pass through the rectangular constraint, but the zero-vector is outside the rectangular

constraint, as shown in the yellow region of Fig. 2. In this case, the double-vector method should be chosen. The number of AVVs passing through the rectangular constraint could be one to three. Case 3 denotes that no active voltage vector (AVV) passes through the rectangular constraint, as depicted in the gray region of Fig. 2. In this situation, the three-vector method is essential for synthesizing the voltage vector at the center of the rectangular constraint.

As a consequence of the above, the number of AVVs passing through the rectangular constraint determines the type of reference voltage vector that is chosen. In the following content, the optimization process of the reference voltage vector will be presented.

1) *Optimization of Single-Vector*: If p_1 and p_2 have opposite signs, and q_1 and q_2 also have opposite signs, then the rectangular constraint contains the origin. In this case, choosing the zero-vector satisfies the independent current constraints and achieves the lowest switching frequency.

2) *Optimal Phase of Double-Vector*: In the stationary coordinate frame, the mathematical expressions for the three AVV lines are defined as follows:

$$\begin{cases} y = 0 \\ y - \sqrt{3}x = 0 \\ y + \sqrt{3}x = 0. \end{cases} \quad (10)$$

According to (9), the four vertices of the rectangular constraint can be calculated as follows:

$$\begin{cases} (x_1, y_1) = \left(\frac{q_1 - p_1}{m - n}, \frac{mq_1 - p_1n}{m - n} \right) \\ (x_2, y_2) = \left(\frac{q_2 - p_1}{m - n}, \frac{mq_2 - p_1n}{m - n} \right) \\ (x_3, y_3) = \left(\frac{q_1 - p_2}{m - n}, \frac{mq_1 - p_2n}{m - n} \right) \\ (x_4, y_4) = \left(\frac{q_2 - p_2}{m - n}, \frac{mq_2 - p_2n}{m - n} \right) \end{cases} \quad (11)$$

where (x_1, y_1) is the coordinate of the point a , which is the intersection point between E_{d1} and E_{q1} . (x_2, y_2) is the coordinate of the intersection point b between E_{d1} and E_{q2} . (x_3, y_3) is the coordinate of the point c , which is the intersection point between E_{d2} and E_{q1} . (x_4, y_4) is the coordinate of the point d which is the intersection point between E_{d2} and E_{q2} .

First, substitute (11) into (12) to determine which AVV lines cross the rectangular constraint. Then select an AVV from them and use its phase as the phase of the reference voltage vector

$$\begin{cases} L_1 = \bigoplus_{i=1}^4 \text{sgn}(y_i) \\ L_2 = \bigoplus_{i=1}^4 \text{sgn}(y_i - \sqrt{3}x_i) \\ L_3 = \bigoplus_{i=1}^4 \text{sgn}(y_i + \sqrt{3}x_i) \end{cases} \quad (12)$$

where $\text{sgn}(x)$ is the sign function. When $x \geq 0$, $\text{sgn}(x) = 1$. When $x < 0$, $\text{sgn}(x) = 0$.

TABLE I
PHASE OPTIMIZATION OF REFERENCE VOLTAGE VECTOR

	$L_1 = 1,$ $L_2 = 0,$ $L_3 = 0$	$L_1 = 0,$ $L_2 = 1,$ $L_3 = 0$	$L_1 = 0,$ $L_2 = 0,$ $L_3 = 1$	$L_1 = 1,$ $L_2 = 1,$ $L_3 = 0$	$L_1 = 0,$ $L_2 = 1,$ $L_3 = 1$	$L_1 = 1,$ $L_2 = 0,$ $L_3 = 1$	$L_1 = 1,$ $L_2 = 1,$ $L_3 = 1$
Sector	I and IV	III and IV	I and II	IV and V	II and III	V and VI	I and II and VI
SAVV	V_1	V_4	V_2	V_5	V_3	V_6	AVV closest to center point

If $L_i = 1$, it means that the corresponding AVV line crosses the rectangular constraint. However, for each AVV line, two voltage vectors with a phase difference of π can be chosen. Based on the sector of the center point of the rectangular constraint, it is easy to determine the locations of the rectangular constraint and the intersections between the rectangular constraint and AVV line. Moreover, when the rectangular constraint intersects two AVV lines, to further minimize the switching frequency, it is preferable to choose AVV resulting in fewer switching actions.

When the rectangular constraint intersects three AVV lines (i.e., $L_1 = 1$, $L_2 = 1$, and $L_3 = 1$), AVV closest to the center of the rectangular constraint can be directly chosen.

Based on the analysis above, the specific method for selecting the AVV used for optimization, which is denoted as selected active voltage vector (SAVV), is shown in Table I.

3) *Optimal Amplitude of Double-Vector*: Once SAVV is selected, the following method is presented to optimize the amplitude of the reference voltage vector. To reduce the steady-state current ripple, the midpoint of the two intersection points between the SAVV line and the rectangular constraint is used to optimize the amplitude of the reference voltage vector. The first step is to determine which two edges of the rectangular constraint intersect with the SAVV line. Then, calculate the intersection points between the SAVV line and these two edges. Consequently, the midpoint of these two intersection points can be obtained to optimize the amplitude of the reference voltage vector.

After determining SAVV according to Table I, substitute (11) into (13)

$$K = \text{sym}(a) + \text{sym}(b) + \text{sym}(c) + \text{sym}(d) \quad (13)$$

$$\text{where } \text{sym}(v) = \begin{cases} \text{sgn}(y_i) & \text{SAVV} = V_1 \text{ or } V_4 \\ \text{sgn}(y_i - \sqrt{3}x_i) & \text{SAVV} = V_2 \text{ or } V_5 \\ \text{sgn}(y_i + \sqrt{3}x_i) & \text{SAVV} = V_3 \text{ or } V_6 \end{cases}$$

“v” is the vertex (a, b, c, or d) of rectangular constraint, and y_i and x_i are the coordinates of the vertex.

All possible values for K are 1, 2, and 3. When K is 1 or 3, it means that one vertex of the rectangular constraint has an opposite sign to the other three vertices. The three vertices with the same sign and another vertex with an opposite sign are located on opposite sides of the SAVV line. Therefore, the two adjacent

TABLE II
GENERAL EQUATION OF $u_\alpha^*(k)$

		$K = 2$	
		$\text{sym}(a) = \text{sym}(b)$	$\text{sym}(a) \neq \text{sym}(b)$
$u_\alpha^*(k)$	$V_1 \text{ or } V_4$	$\frac{-V_q}{Q_1}$	$\frac{V_d}{D_1}$
	$V_2 \text{ or } V_5$	$\frac{V_q}{Q_2}$	$\frac{V_d}{D_2}$
	$V_3 \text{ or } V_6$	$\frac{-V_q}{Q_3}$	$\frac{V_d}{D_3}$

$$\text{Note: where } \begin{cases} D_1 = 2T\alpha_d \cos \theta_r \\ D_2 = 4T\alpha_d \sin(\pi/6 + \theta_r) \\ D_3 = 4T\alpha_d \sin(\pi/6 - \theta_r) \end{cases}, \begin{cases} Q_1 = 2T\alpha_q \sin \theta_r \\ Q_2 = 4T\alpha_q \sin(\pi/3 - \theta_r) \\ Q_3 = 4T\alpha_q \sin(\pi/3 + \theta_r) \end{cases}$$

$$\begin{cases} N_1 = (-1)^{\frac{K-1}{2}}(2(\text{sym}(c) + \text{sym}(d)) - K) \\ N_2 = (-1)^{\frac{K-1}{2}}(2(\text{sym}(b) + \text{sym}(d)) - K) \end{cases}$$

edges of the vertex with an opposite sign intersect the SAVV line, forming rectangle U as shown in Fig. 2. When K is 2, it indicates that there are two vertices of the rectangular constraint on each side of the SAVV line, as shown in rectangles V and W in Fig. 2. If vertices “a” and “b” are both located on the same side of the SAVV line (i.e., $\text{sym}(a) = \text{sym}(b)$), the nonadjacent edges of the rectangular constraint that intersect the SAVV line are E_{q1} and E_{q2} , as shown in rectangle V in Fig. 2. Otherwise, the two nonadjacent edges are E_{d1} and E_{d2} , as shown in rectangle W in Fig. 2. The optimal reference voltage vector amplitude corresponds to the midpoint between the two intersection points. The detailed calculations for $u_\alpha^*(k)$ are shown in Table II.

In this case, the expression for $u_\beta^*(k)$ at the midpoint between the two intersection points can be represented as follows:

$$u_\beta^*(k) = \begin{cases} 0 & \text{SAVV} = V_1 \text{ or } V_4 \\ \sqrt{3}u_\alpha^*(k) & \text{SAVV} = V_2 \text{ or } V_5 \\ -\sqrt{3}u_\alpha^*(k) & \text{SAVV} = V_3 \text{ or } V_6 \end{cases} \quad (14)$$

4) *Optimization of Three-Vector*: When all AVV lines do not pass through the rectangular constraint, such as the grey rectangle in Fig. 2, the center point of the rectangular constraint is chosen as the reference voltage vector, which is synthesized by the three-vector method. The coordinates of the center point are given as follows:

$$\begin{cases} u_\alpha^*(k) = \frac{x_1 + x_4}{2} \\ u_\beta^*(k) = \frac{y_1 + y_4}{2} \end{cases} \quad (15)$$

It is worth noting that, in this method, choosing the midpoint of the intersections between the rectangular constraint and the SAVV may not minimize $J_d + J_q$ in some cases. However, this could prevent J_d or J_q from reaching their allowable maximum values, thereby avoiding higher current and torque ripple.

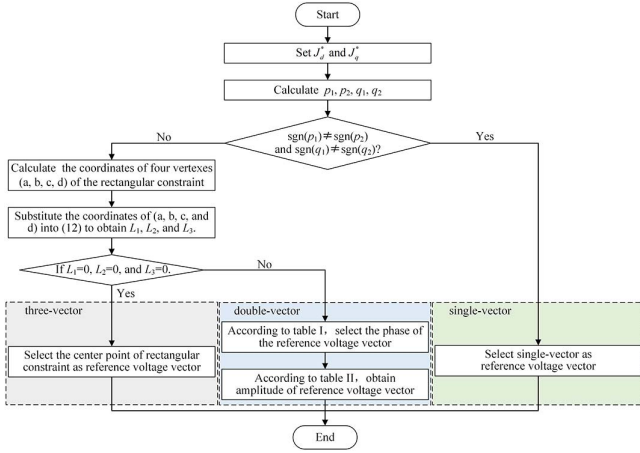


Fig. 3. Flowchart of ICF-MPC.

The flowchart of the proposed ICF-MPC is shown in Fig. 3. Fig. 4 shows the block diagram of SMPMSM drive based on the proposed ICF-MPC.

IV. EXPERIMENTAL RESULTS

To validate the control performance of the proposed ICF-MPC, experimental research was conducted on an SMPMSM test bench, as shown in Fig. 5. The dynamometer employed in the experiments is a 2.2 kW induction motor, which is controlled using an ABB frequency converter, operating in speed mode. The nominal parameters of the tested SMPMSM are shown in Table III. The rotor position is obtained from a resolver, and the three-phase current is measured using Hall current sensors. All four methods, duty control model predictive current control with pulse pattern F (DC-MPC) [25], optimal solution for model predictive current control (OS-MPC) [26], dead beat predictive control (DPC), and the proposed ICF-MPC, are executed on the dSPACE/DS1007 experimental platform. All four methods are based on the ultra-local model to avoid the impacts of parametric uncertainties and inverter nonlinearity. The scaling factors are set as $\alpha_d = \alpha_q = 1/L_s = 1000$. The inverter is configured with a dead time 2 μ s. Consistent testing platform, data measurement equipment, and control parameters are applied across all experiments to ensure the comparative fairness of different methods. The sampling period and control period of OS-MPC, DPC, and ICF-MPC are 100 μ s, whereas DC-MPC is set as 70 μ s to maintain a similar switching frequency to the other three methods. OS-MPC and DPC both use the same DPWM method, so their switching frequency is 6.67 kHz. The d -axis reference current is set to 0 A to achieve maximum torque per ampere control in all experiments.

A. Independent Current Constraint Test

To verify the effectiveness of the proposed ICF-MPC for dq -axes current independent constraints, J_d^* and J_q^* in the experiments are set to different values to prove the ability of J_d^* and J_q^*

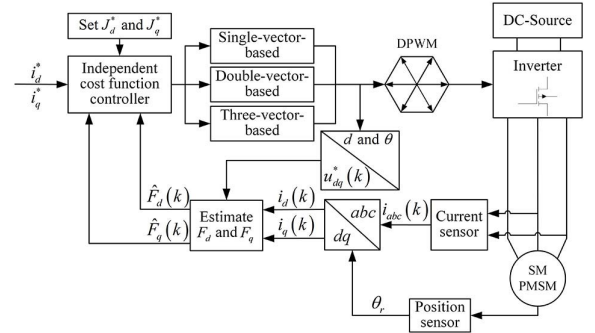


Fig. 4. Control block diagram of ICF-MPC for SMPMSM drive.

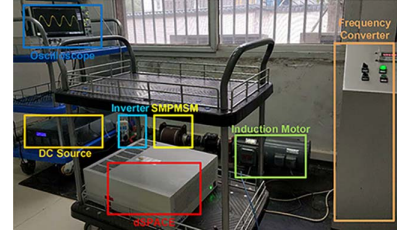


Fig. 5. Experimental bench.

TABLE III
NOMINAL PARAMETERS OF SMPMSM

Name	Value
Rated torque	13 N·m
Rated speed	500 rpm
Rated current	19 A
Pole pairs	12
Stator inductance	1 mH
Stator resistance	0.0957 Ω
PM flux linkage	0.027 Wb
Moment of inertia	0.01015 kg·m ²

in the flexibly and independently controlling dq -axes current ripples.

In the first experimental test, only J_q^* is varied, while J_d^* is kept constant at 0.4². The values of J_q^* are set to 0.4², 0.6², 0.8², and 1², respectively. The q -axis reference current is 20 A, and the rotor mechanical speed is 100 rpm. The corresponding experimental results are presented at the top of Fig. 6. On the one hand, as J_q^* increases, the q -axis steady-state current ripple also increases. However, it remains within the specified current constraint. On the other hand, since J_d^* remains unchanged, the d -axis steady-state current ripple remains essentially constant throughout the entire process, not varying with the value of J_q^* .

Subsequently, the experimental condition is changed so that J_q^* is kept constant, while J_d^* was sequentially set to 0.4², 0.6², 0.8², and 1². The experimental result is depicted at the bottom of Fig. 6. It shows that the d -axis steady-state current ripple increases with the increase of J_d^* , and the q -axis steady-state current ripple remains essentially constant. The above experimental

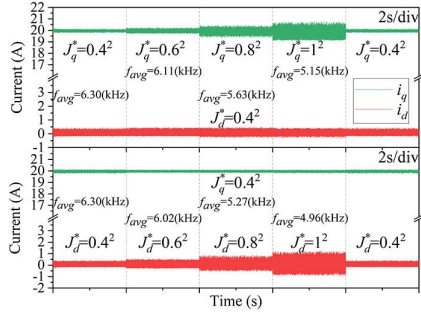


Fig. 6. Steady-state current response of the proposed ICF-MPC.

results demonstrate that the proposed ICF-MPC can separately constrain the dq -axes currents to achieve independent control of the dq -axes steady-state currents.

B. Dynamic Response Comparison

To validate the dynamic response of the proposed ICF-MPC, a comparative experiment is conducted. Since the electromagnetic torque of SMPMSM is directly related to the q -axis current, J_q^* is set to 0 and 0.15^2 to reduce torque ripple, while J_d^* is set to 0.5^2 , 1.5^2 , and 0.2^2 to decrease inverter switching frequency, respectively. The speed of SMPMSM is regulated by a dynamometer and sequentially set to 100 and 400 rpm. A step change of 10 A is applied to the q -axis reference current at 0.01 s.

The experimental results of the four control methods are depicted in Fig. 7. From the comparison of the dynamic response of the four methods in Fig. 7, it can be seen that the proposed ICF-MPC has a similar dynamic response to DC-MPC, OS-MPC, and DPC. Moreover, Fig. 7(g)–7(l) shows the dynamic response of the proposed ICF-MPC when setting three different J_d^* and J_q^* . They demonstrate the same dynamic response. At the same time, Fig. 7(g)–7(l) further illustrates the capability of ICF-MPC in flexibly controlling the dq -axes current performance.

C. Verification of the Multiobjective Optimization

To verify the multiobjective optimization effectiveness of the proposed ICF-MPC, we conducted experimental comparisons between DC-MPC, OS-MPC, DPC, and ICF-MPC. To more clearly compare the four methods, J_d^* and J_q^* of the proposed ICF-MPC are set to 0.2^2 and 0.15^2 , respectively, in the following experiments. Because the dq -axes current ripples of OS-MPC, DPC, and ICF-MPC are at the same level under these settings. The experimental results are shown in Figs. 8–17. Figs. 8–11 show the dq -axes and a -phase steady-state current response of the four methods under all conditions, respectively. From these figures, it can be seen that DC-MPC has the worst steady-state current performance because it only uses the combination of one AVV and one zero-vector as the reference voltage vector. Furthermore, through the proper design of J_d^* and J_q^* , the proposed ICF-MPC not only achieves a d -axis current ripple similar to OS-MPC but also effectively reduces the q -axis current ripple. In Figs. 12, 13, and 17, C_f is used to represent the comprehensive performance of the different methods in terms of

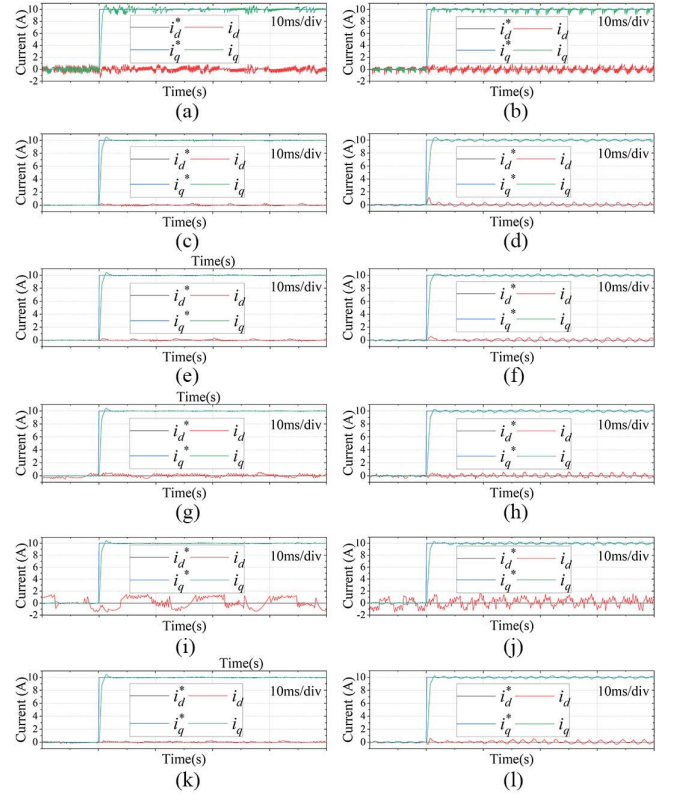


Fig. 7. Dynamic current response of different methods. (a) DC-MPC at 100 rpm/10 A. (b) DC-MPC at 400 rpm/10 A. (c) OS-MPC at 100 rpm/10 A. (d) OS-MPC at 400 rpm/10 A. (e) DPC at 100 rpm/10 A. (f) DPC at 400 rpm/10 A. (g) $J_d^* = 0.5^2$, $J_q^* = 0$ at 100 rpm/10 A. (h) $J_d^* = 0.5^2$, $J_q^* = 0$ at 400 rpm/10 A. (i) $J_d^* = 1.5^2$, $J_q^* = 0$ at 100 rpm/10 A. (j) $J_d^* = 1.5^2$, $J_q^* = 0$ at 400 rpm/10 A. (k) $J_d^* = 0.2^2$, $J_q^* = 0.15^2$ at 100 rpm/10 A. (l) $J_d^* = 0.2^2$, $J_q^* = 0.15^2$ at 400 rpm/10 A.

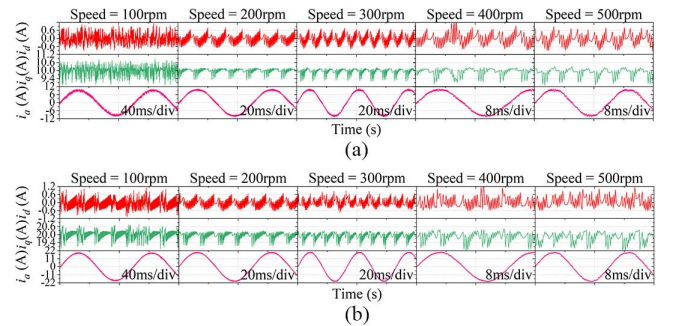


Fig. 8. Steady-state current response of DC-MPC. (a) 10 A. (b) 20 A.

current and switching frequency. C_f is an insightful performance metric [27], characterized as the product of the total harmonic distortion (THD) of the a -phase current and average switching frequency f_{avg} , reflecting the balance between THD and f_{avg} . As illustrated in Fig. 12, the proposed ICF-MPC can significantly reduce C_f compared to DC-MPC under all operations. At low-speed operations, the proposed ICF-MPC reduces C_f by more than 17.31% compared to DC-MPC. Notably, at 100 rpm/10 A, it achieves a significant reduction of 63.65% in C_f . And at 100 rpm/20 A, it delivers a reduction of 59.40%.

Since both DPC and OS-MPC aim to make the practical current in the next period track the reference current, the

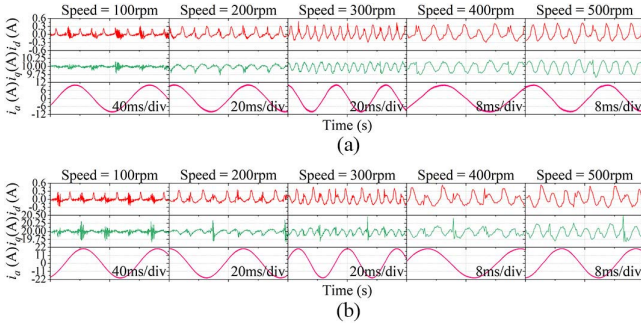


Fig. 9. Steady-state current response of OS-MPC. (a) 10 A. (b) 20 A.

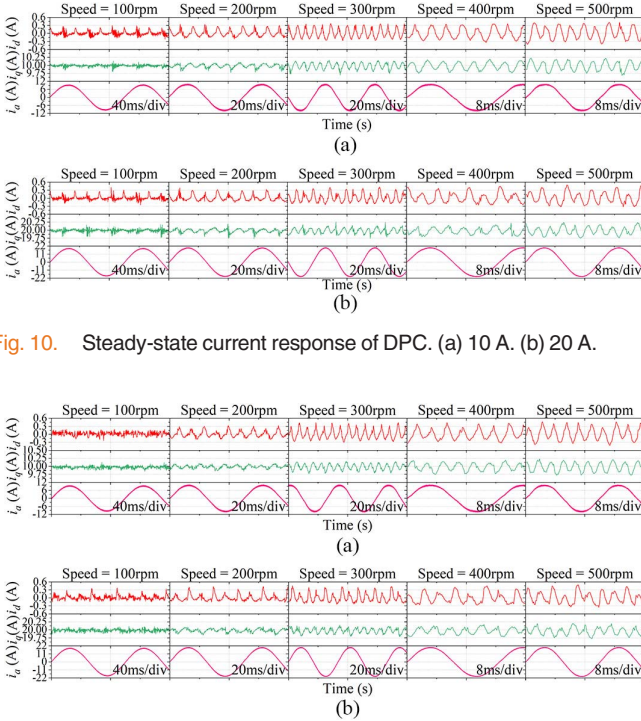


Fig. 10. Steady-state current response of DPC. (a) 10 A. (b) 20 A.

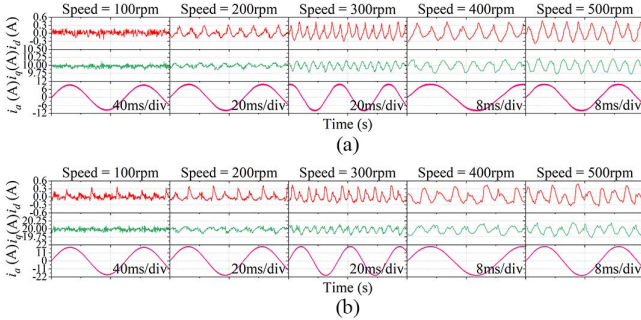


Fig. 11. Steady-state current response of ICF-MPC. (a) 10A. (b) 20A.

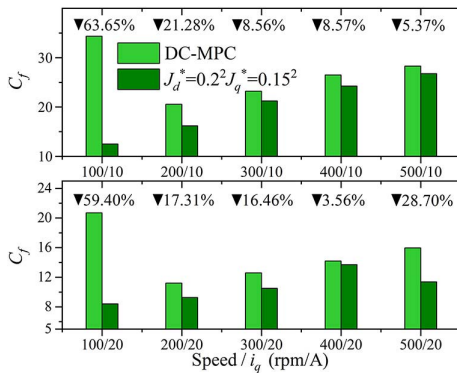


Fig. 12. Comprehensive performance comparison between DC-MPC and ICF-MPC.

experimental results of OS-MPC and DPC can be used as benchmarks. Therefore, we conducted a more detailed comparison between OS-MPC, DPC, and ICF-MPC in Figs. 13–17. Fig. 13 shows the comparison of the comprehensive performance

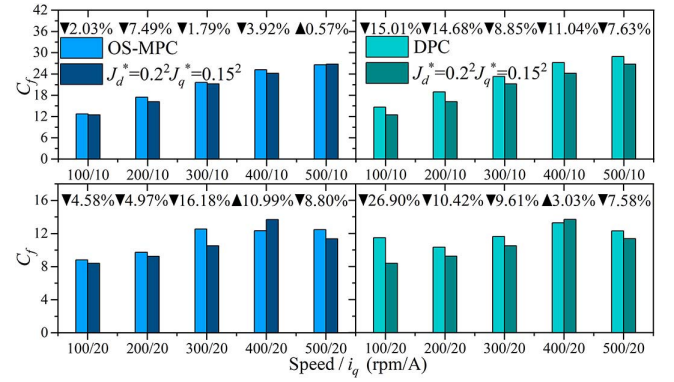


Fig. 13. Comprehensive performance comparison between OS-MPC, DPC, and ICF-MPC.

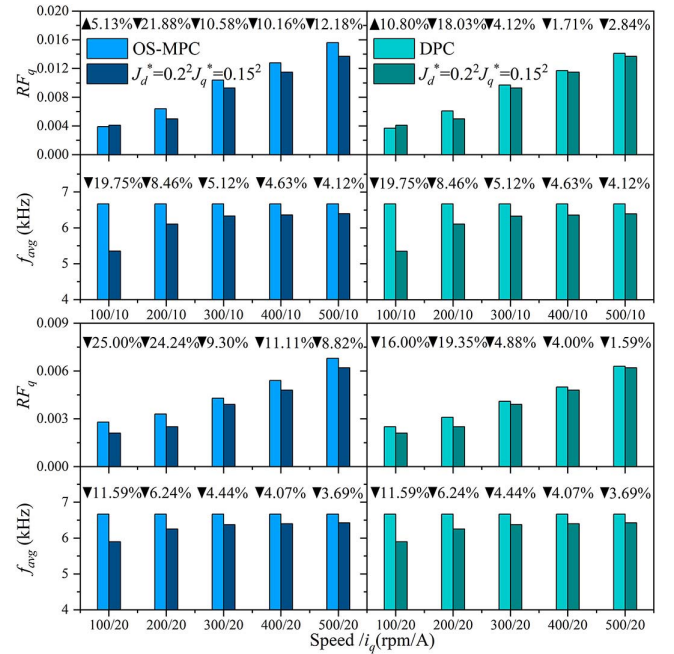


Fig. 14. Comparison of q -axis current ripple factor and switching frequency between OS-MPC, DPC, and ICF-MPC.

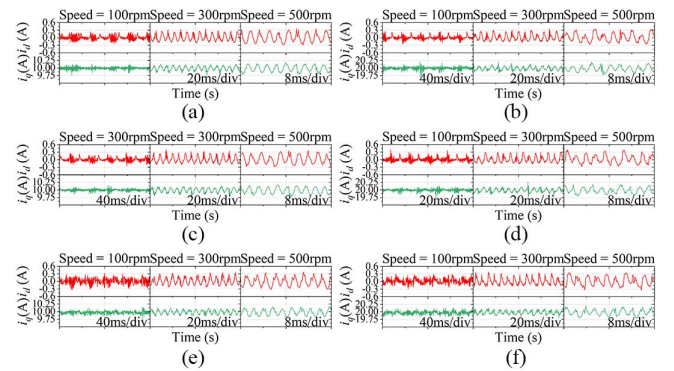


Fig. 15. Steady-state current response of OS-MPC, DPC, and ICF-MPC when $\alpha_d = \alpha_q = 750$. (a) OS-MPC at 10 A. (b) OS-MPC at 20 A. (c) DPC at 10 A. (d) DPC at 20 A. (e) ICF-MPC at 10 A. (f) ICF-MPC at 20 A.

between the three methods. Fig. 14 illustrates the comparison of the q -axis current ripple factor (RF_q) [28] and the average switching frequency (f_{avg}) between the three methods.

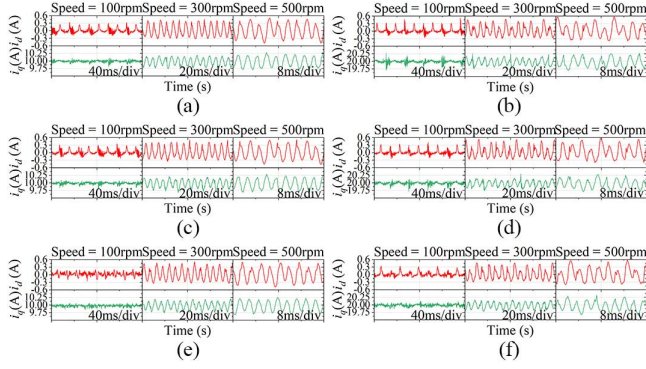


Fig. 16. Steady-state response of OS-MPC, DPC, and ICF-MPC when $\alpha_d = \alpha_q = 1250$. (a) OS-MPC at 10 A. (b) OS-MPC at 20 A. (c) DPC at 10 A. (d) DPC at 20 A. (e) ICF-MPC at 10 A. (f) ICF-MPC at 20 A.

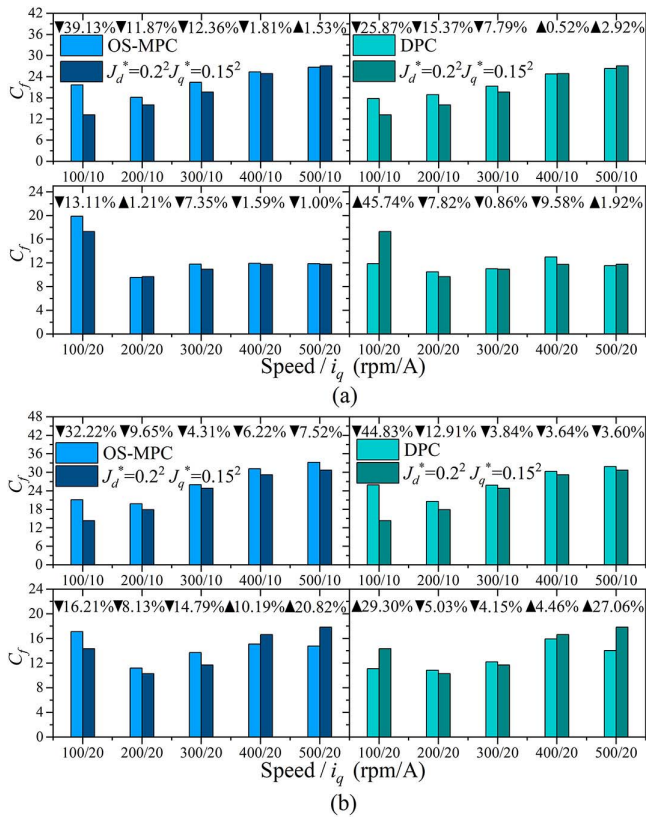


Fig. 17. Comprehensive performance comparison between OS-MPC, DPC, and ICF-MPC under parameter mismatch. (a) $\alpha_d = \alpha_q = 750$. (b) $\alpha_d = \alpha_q = 1250$.

Figs. 15–17 present the performance comparison of the three methods under parameter mismatch conditions. From Fig. 13, it is evident that, compared to OS-MPC, the proposed ICF-MPC achieves better overall performance in all operations except 500 rpm/10 A and 400 rpm/20 A. The proposed ICF-MPC only performs worse than DPC at 400 rpm/20 A, while it performs better than DPC in all other conditions. These results show a significant enhancement in the comprehensive performance of the stator current and inverter switching frequency. From Fig. 14, it can be seen that, compared to DPC and OS-MPC, the proposed ICF-MPC achieves simultaneous reductions in RF_q and f_{avg} in all operations except 100 rpm/10 A. Moreover, at 100 rpm/10

TABLE IV
TURNAROUND TIME OF DIFFERENT METHODS

	DC-MPC	OS-MPC	DPC	ICF-MPC
Turnaround time (μ s)	19.0	19.0	18.6	19.5

A, compared to OS-MPC, the proposed ICF-MPC's RF_q increased by 5.13% but achieved a 19.75% reduction in f_{avg} . Notably, at 100 rpm/20 A, the proposed ICF-MPC achieves a 25% reduction in RF_q and an 11.59% reduction in f_{avg} compared to OS-MPC. The proposed ICF-MPC can also achieve similar results compared to DPC at 100 rpm/10 A and 100 rpm/20 A.

Since only α_d and α_q in the ultra-local model are related to SMPMSM parameters, we set α_d and α_q to 0.75 times (750) and 1.25 times (1250) the initial value to compare the performance of OS-MPC, DPC and ICF-MPC under the parameter mismatch conditions. As shown in Figs. 15 and 16, when $\alpha_d = \alpha_q = 750$, ICF-MPC has d -axis current ripple that either larger than or similar to that of OS-MPC and DPC. When $\alpha_d = \alpha_q = 1250$, the d -axis current ripple of ICF-MPC is smaller than or similar to that of OS-MPC and DPC. However, in most cases, the q -axis current ripple of ICF-MPC is lower compared to both OS-MPC and DPC. From Fig. 17(a), it can be seen that, in ten operating conditions, ICF-MPC performs better than OS-MPC in eight conditions and better than DPC in six conditions when $\alpha_d = \alpha_q = 750$. Particularly, at 100 rpm/10 A, ICF-MPC achieves a 39.13% reduction in C_f compared to OS-MPC. From Fig. 17(b), it can be seen that when $\alpha_d = \alpha_q = 1250$, under 10 operating conditions, ICF-MPC outperforms both OS-MPC and DPC in seven conditions, except 100 rpm/20 A, 400 rpm/20 A, and 500 rpm/20 A. Moreover, at 100 rpm/20 A, ICF-MPC also performs better than OS-MPC. Specifically, at 100 rpm/10 A, compared to OS-MPC, ICF-MPC reduces C_f by 32.22%, and compared to DPC, this reduction is 44.84%.

D. Calculation Burden

The computation burdens of DC-MPC, OS-MPC, DPC, and ICF-MPC are compared. Due to the software restriction of dSPACE, only the turnaround time of the control algorithms can be measured. We present the turnaround time of the four methods in Table IV. As can be seen from Table IV, the computational burden of ICF-MPC shows a negligible increase compared to DC-MPC, OS-MPC, and DPC. This main reason is that the proposed ICF-MPC only involves additional judgement conditions, without incorporating complex computation.

V. CONCLUSION

In this article, an independent cost function model predictive current control method is proposed to achieve the trade-OFF optimization among d -axis current performance, q -axis current performance, and the inverter switching frequency. The proposed method incorporates independent dq -axes current constraints, enabling the separate control of d -axis and q -axis current performance, in accordance with the specification of requirements of SMPMSM drive. Then, with the flexible current

constraints, the proposed ICF-MPC utilizes the hybrid voltage vector to significantly reduce the switching frequency without the weighting factors. Experimental results indicate that the dynamic response and computational load are comparable to those of conventional MPC methods. In pursuit of more effective reduction of the inverter switching frequency, further work could explore optimizing the application sequence of voltage vectors as an alternative to DPWM. Additionally, when incorporating more optimization objectives, further research is required to refine the proposed method.

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